

Beyond Fashion Vocabulary: Spatial Style Navigation Through Dual-Layer Style-Structural Mapping in XR

Wen Li ^{*}Shiyun Li [†]Xiaoxuan Zhao [‡]Mengting Lai [§]Pasquale Cascarano ^{||}Gustavo Marfia ^{||}

University of Bologna, Italy

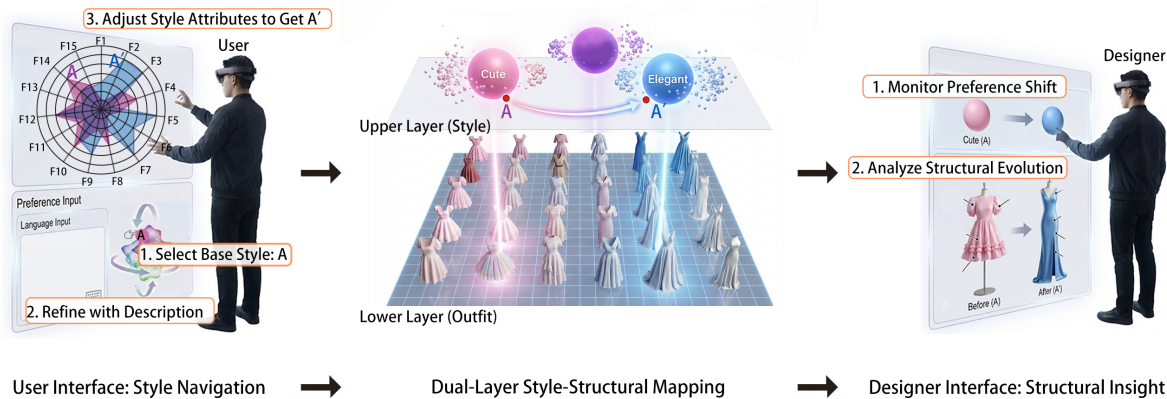


Figure 1: **System workflow.** **Left:** User selects base style A, refines with natural language, and adjusts via the semantic radar to reach A'. **Center:** Dual-layer architecture—upper layer (affective manifold) and lower layer (structural outfit database)—with UMAP mapping between them. **Right:** Designer monitors preference shifts and analyzes structural evolution from A to A'. Spatial navigation eliminates vocabulary barriers while providing behavioral insights.

ABSTRACT

We argue that forcing users to describe visual aesthetics with text represents a fundamental architectural flaw in digital fashion systems. We propose a dual-layer framework as a conceptual alternative that reframes style discovery from a linguistic task to a spatial navigation experience in eXtended Reality (XR). The system's upper layer maps fashion styles into a navigable 3D manifold based on 15 affective dimensions derived from Kansei Engineering. The lower layer contains 3,000 designer-curated outfit configurations with detailed structural annotations. Users explore the style landscape through spatial movement and refine their selections through a Dual-Layer Style-Structural Mapping interface radar, which maps real-time affective adjustments to corresponding garment structures. We contend that this spatialized approach bypasses semantic barriers and provides designers with objective behavioural data through recorded navigation trajectories. We discuss design details, interaction methods, and key challenges, including cultural bias, interpretability, and scalability.

Index Terms: Extended Reality, Digital Fashion, Vocabulary Gap, Spatial Navigation, Human-AI Co-Creation, Kansei Engineering

*e-mail: wen.li7@unibo.it

†e-mail: shiyun.li3@unibo.it

‡e-mail: xiaoxuan.zhao2@unibo.it

§e-mail: mengting.lai@unibo.it

||e-mail: pasquale.cascarano2@unibo.it

||e-mail: gustavo.marfia@unibo.it

1 INTRODUCTION

Consider a user looking for a specific vibe, perhaps something “playfully retro but not too costume-like”. In traditional interfaces, this intent hits a wall of dropdown menus and rigid keywords. The interface demands a vocabulary that the user doesn't possess [2]. We argue that this creates a dual problem: users struggle to articulate style preferences using professional terminology, and systems cannot capture the subjective, semantically ambiguous nature of fashion esthetics [28, 9]. We call this the **Vocabulary Gap**, not merely a usability issue, but an architectural misalignment between human cognition and system design. While Generative Artificial Intelligence (GenAI) offers partial relief, it often lacks transparency, leaving users unable to discern which prompt elements triggered which outcomes. It is a problem of opacity rather than capability.

Previous research has attempted to bridge this gap through Kansei Engineering and semantic embeddings [25]. However, these approaches typically treat styles as discrete categories rather than continuous spectra, providing limited mechanisms for iterative refinement [34]. Consequently, users remain unable to explore the relational nuances between styles or systematically adjust specific aesthetic dimensions.

We propose that XR offers a path forward, but only if we move beyond 2D paradigms. Rather than treating fashion styles as discrete database entries, we advocate for visualizing them as a continuous, navigable landscape where spatial proximity encodes aesthetic similarity. Our position is straightforward: by externalizing the mental model of style into a physical, navigable space, we reduce the cognitive load of retrieval.

2 RELATED WORK

Our work builds upon three research streams: (1) Quantified style representation in computational fashion, (2) Semantic-based re-

trieval navigation mechanisms, (3) and Human-AI collaborative design systems.

2.1 Quantified Representation and Structural Modeling

Fashion style is inherently subjective and semantically ambiguous, shaped by diverse cultural contexts and individual designer experiences [24]. We argue this subjectivity hinders computational systems that rely on structured, quantifiable representations for retrieval. [4]. Zhang employed the Semantic Differential Method and Triangular Fuzzy Number models to bidirectionally quantify style dimensions such as “Elegant Neutral” and “Ethnic Futuristic”, establishing foundations for style databases and recommendation systems [33]. These approaches model style imagery based on affective word spaces, translating subjective impressions into measurable parameters.

While Kansei Engineering has successfully modeled affective dimension translating subjective impressions like “Elegant” or “Neutral” into measurable parameters, current research remains limited by its unidirectional nature. Existing models typically map isolated structural elements (e.g., collar types) to specific vocabulary without analyzing how styles relate to each other across a full semantic space. They lack a systematic exploration of similarity and overlap, preventing the meaningful clustering required for intuitive navigation.

Recent work has begun to address this through semantic distance calculations. Adaptive loss functions map fashion images into embedding spaces where semantically similar samples cluster together, enabling precise retrieval and ranking [31]. Multilevel text-to-outfit retrieval frameworks extract semantic representations at the item, style, and outfit levels, introducing clustering of K-means Clustering to achieve inter-class differentiation in the semantic space of style [13]. Yet we observe that these approaches still treat styles as discrete clusters rather than points on a continuous, navigable spectrum.

Deep learning has advanced automatic style feature extraction and classification. Transfer learning-optimized residual networks measure style similarity between images [14]. Design Issue Graphs (DIG) transform local attribute information into global semantic graphs, improving consistency and robustness in style recognition [32]. Improved Bilinear-Convolutional Neural Network (CNN) models with spatial attention mechanisms achieve 76.4% accuracy on the FashionStyle14 dataset, demonstrating adaptability to fine-grained style recognition [18]. Few-shot learning methods reach 0.9475 accuracy in 5-way, 1-shot tasks for multi-brand style recognition, showing strong generalization even with sparse samples [11].

Although deep learning approaches, such as transfer learning and bilinear-CNNs, have improved fine-grained style recognition, they treat styles as discrete classification tasks rather than points on a continuous spectrum. While these systems can recognize individual styles, they do not expose the relational structure that would enable users to understand how to navigate incrementally between aesthetics.

2.2 Semantic Retrieval and High-Dimensional Space Navigation

Keyword search fundamentally cannot capture complex visual intentions. This limitation has driven research toward semantic retrieval based on embedding spaces [22], where fashion images and text map into shared high-dimensional vector spaces with distance metrics reflecting semantic similarity [6].

Multi-level retrieval frameworks enhance precision by learning semantic representations across item, style, and outfit levels, and employing K-means-based clustering to encourage inter-class separation within the style embedding space [28]. These methods

optimize mapping relationships through adaptive loss functions, achieving ranked retrieval results.

However, these systems remain transactional: users cannot explore the retrieval space or visualize relationships between style options. We argue that XR addresses this limitation through spatial affordances. By representing styles as points in a navigable 3D space, XR would enable users to perceive aesthetic relationships holistically and refine preferences through embodied spatial movement.

2.3 Human-AI Co-Creativity and Logic Transparency

Generative AI has shifted design tools from drawing assistance toward Human-AI co-creativity. Effective co-creative systems should grant users fine grained control over generative results, positioning them as active collaborators rather than passive prompt inputters [23, 29]. Current research enhances intent expression through multimodal inputs: sketches, text, poses [1]. However, most generative fashion tools still suffer from a gulf of execution: users possess abstract affective intent (e.g., “fragmented feeling”) but cannot precisely control the specific structural parameters the AI generates [8].

We argue the gap widens when AI decision-making processes remain opaque: users cannot understand why the model recommended a particular outfit, nor can they correct biased interpretations of specific styles. [7]. This opacity is particularly problematic in fashion, where users often recognize preferences only upon seeing options rather than articulating them in advance. The vocabulary gap is thus compounded by a transparency gap: users lack both the terminology to describe their intent and visibility into how their inputs map to system outputs.

3 DESIGN GOALS

To address the limitations identified above, we propose a framework guided by three design goals:

DG 1: Spatial Style Exploration. We propose moving beyond filtered lists to treat style as a continuous landscape. By constructing a 3D Style Manifold where spatial proximity correlates with aesthetic similarity, users would be able to perceive relationships holistically and grasp the full aesthetic spectrum before making selections.

DG 2: Transparent Style Composition. To counter opacity in recommendation systems, users should be able to understand the logic underlying specific recommendations. We propose a Semantic Radar that decomposes style points into weighted dimensions (e.g., Simple-Complex), revealing the compositional logic of each style.

DG 3: Support Iterative Refinement. We argue that interaction should be manipulative, not merely iterative. Through a **Dual-Layer Style-Structural Mapping** interface, users would adjust affective parameters on the radar while the system retrieves structurally aligned garment configurations in real-time.

4 DUAL-LAYER STYLE-STRUCTURAL MAPPING

We propose a solution that separates affective navigation from structural retrieval while maintaining a transparent, adjustable link between them.

4.1 Upper Layer: Multi-Dimensional Affective Quantification

The upper layer employs Kansei Engineering to map user’ subjective esthetic intent onto a 15 Dimensional affective space organized into five facets [17]: Gender, Age, Fashionability, Composition and Temperament (Table 1). In our prior work [17], ten professional fashion designers independently scored each style on all 15 dimensions using 7-point Likert scales. This approach aligns

Table 1: 15 Adjective Pairs across 5 Facets

Facet	Adjective Pairs		
Gender	F1: Feminine—Masculine	F6: Sweet—Cold	F11: Elegant—Rugged
Age	F2: Youthful—Mature	F7: Cute—Sophisticated	F12: Nostalgic—Modern
Fashionability	F3: Niche—Mainstream	F8: Luxurious—Plain	F13: Unique—Ordinary
Composition	F4: Simple—Complex	F9: Colorful—Achromatic	F14: Classic—Trendy
Temperament	F5: Light—Steady	F10: Sexy—Conservative	F15: Romantic—Solemn

with established research using Kansei to construct affective semantic spaces and establish quantitative relationships with design elements [5, 19].

Building on prior work identifying 54 distinct outfit styles (Figure 2), we perform dual encoding on each style. Each configuration receives an Affective Intention Scoring Vector (t_i) from expert Kansei scoring [17] and a Semantic Embedding Vector (b_i) extracted via pre-trained Bidirectional Encoder Representations from Transformers (BERT) models [12] (Figure 3). This parallel modeling of affective and semantic features before fusion resonates with recent advances in multimodal affective analysis [27, 20]. We concatenate these vectors ($a_i = [t_i; b_i]$) to bridge explicit semantics captured by language models and implicit nuances perceived by human experts.



Figure 2: 54 common fitting styles.

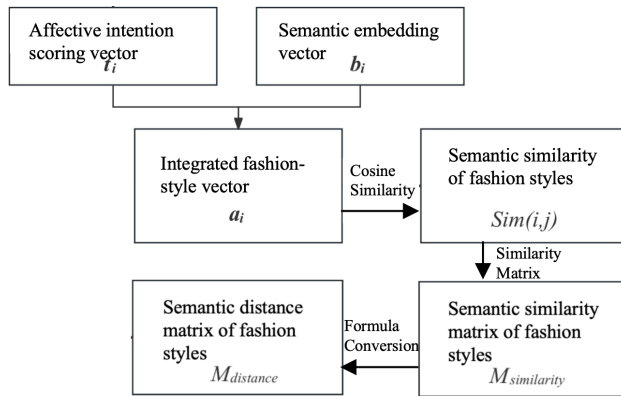


Figure 3: Dual encoding pipeline. Each outfit receives an Affective Intention Scoring Vector (t_i) from Kansei scoring and a Semantic Embedding Vector (b_i) from Bidirectional Encoder Representations from Transformers (BERT) models. Concatenation produces Integrated Fashion-Style Vectors (a_i). Cosine similarity generates a Semantic Similarity Matrix ($M_{similarity}$), converting to Semantic Distance Matrix ($M_{distance}$) for affective-to-structural mapping.

To enable spatial navigation in XR, we employ Uniform Mani-

fold Approximation and Projection (UMAP) to project these high-dimensional vectors into a navigable 3D environment [6]. UMAP has demonstrated effectiveness in preserving structure across diverse datasets and aligning geometric proximity with perceptual similarity in images, artworks, and word embeddings [21, 16]. Drawing on these precedents, we anticipate Euclidean distance in the XR space would approximate aesthetic similarity across semantic and affective dimensions.

4.2 Lower Layer: Structural Manifold

The lower layer would leverage a database of expert-validated outfit configurations, systematically annotated to establish a concrete design space [17]. Building on prior work that developed 3,000 annotated configurations, we propose a granular annotation schema that differs from generic tagging approaches.

Tops would be decomposed into five components: Style, Body, Collar, Sleeve, and Pattern. Annotations would capture both morphological and color features. Morphological features include structural elements such as placket, shoulder, and dart, as well as dimensional parameters such as body ease, garment length, and waistline position. Color features would be annotated separately for Body, Collar, Sleeve, and Pattern components.

Bottoms would be divided into five regions: Waist, Waist-to-Hip, Hip, Hip-to-Hem, and Hem. Each region would receive morphological annotations covering structural elements such as pockets, horizontal divisions, and gathering, along with dimensional parameters such as waistline height and distance-from-body. Color annotations would be provided for each region.

Accessories would receive annotations for body attributes including style, attachment location, and wearing method, as well as pattern attributes including style and placement location. Separate color features would be annotated for each accessory.

This granularity is designed to enable the system to map abstract adjustments to concrete constraints. If a user tunes their preference toward “Complex”, the system does not guess; it retrieves configurations tagged with correlated attributes like asymmetric cuts or layered elements.

4.3 Bridging User Intent and Designer Insight

Standard generative tools suffer from a “Gulf of Evaluation”: users cannot discern why results were generated [8]. We propose a Semantic Radar interface to make affective to structural mapping transparent and interactive, drawing on research demonstrating that structured preference displays improve user trust [10] and that multi-attribute visual controls enable preference refinement through visual-spatial manipulation [26].

User Perspective: When selecting a point in the 3D manifold, users would see its affective composition visualized across all 15 dimensions. Direct manipulation of radar vertices extending the “Cute” axis, for instance would trigger immediate retrieval of aligned outfits (Figure 4). This aligns with work showing that editable feature profiles make recommendations more scrutable [26]. We anticipate this visual-spatial manipulation would enable preference refinement without verbalizing abstract intent, addressing the vocabulary gap through embodied interaction.

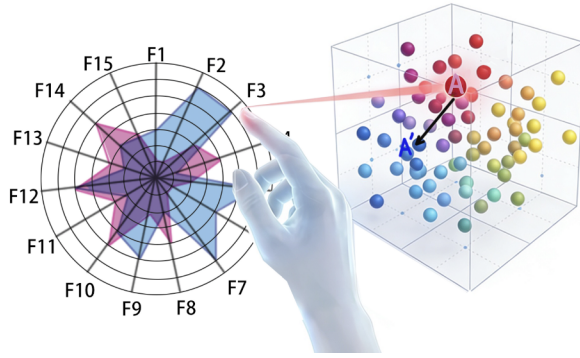


Figure 4: Semantic Radar workflow: Users explore the 3D Style Manifold, select a point, adjust its radar representation, and receive updated outfit suggestions. The system logs navigation trajectories for designer analysis.

Designer Perspective: The system would log entire exploration trajectories, which styles users approach, which dimensions they adjust, and how preferences evolve. This is grounded in sequential recommender research showing interaction trajectories encode dynamic preferences more richly than final choices [3, 30, 15], revealing how users navigate rather than what they claim to prefer.

This bidirectional transparency is designed to create dual value. Users would discover outfits aligned with preferences they struggle to articulate verbally. Designers would decode those preferences at both abstract levels which affective dimensions drive exploration and concrete levels which structural elements resonate, enabling data-driven refinement of the lower layer’s outfit configurations.

5 CRITICAL CHALLENGES AND FUTURE WORK

Our framework faces four fundamental obstacles requiring transparent acknowledgment and community collaboration.

5.1 Cultural Bias in Affective Dimensions

Our 15 affective adjective pairs encode culturally specific aesthetics. The term “Elegant” emphasizes simplicity in Western fashion but might involve intricate embroidery in Chinese traditions. Projecting these culturally-situated judgments into a shared 3D space risks creating a manifold that feels natural to some users while alienating others whose aesthetic frameworks do not align with our dimensions. The lower layer compounds this issue. Expert-curated configurations may disproportionately represent Western garment structures such as collared shirts, while underrepresenting garments such as Hanfu or Sarees. Users from underrepresented cultures may find their preferences map to sparse, poorly-populated regions.

Open Question 1: Should we construct culturally specific sub-manifolds, or can a single global manifold accommodate different aesthetic philosophies?

5.2 Radar Interpretability

Assuming the manifold adequately represents diverse aesthetics, users must still interpret affective adjustments. Whether users understand what increasing “Cute” from 1 to 7 means in practice remains unclear. Users might adjust parameters without grasping how those changes translate to garment structures.

Open Question 2: What contextual information helps users build accurate mental models?

5.3 Temporal Drift

Even with interpretable controls, the manifold must adapt to fashion’s evolution. Styles labeled trendy today become dated rapidly.

Our manifold captures a temporal snapshot, yet expectations shift continuously. Adding configurations reflecting emerging trends may cause them to cluster incoherently. Retraining to accommodate new styles may disrupt spatial organization, invalidating users’ learned navigation strategies.

Open Question 3: How can we maintain manifold stability while incorporating evolving trends without disrupting users’ spatial mental models?

5.4 Scalability

All previous challenges assume sufficient data coverage. Our approach currently relies on 3,000 expert-curated configurations with approximately 100 structural tags each, totaling 300,000 manual annotations. Sparse coverage breaks continuous navigation as users encounter gaps in underrepresented regions. Automated tagging risks corrupting the affective-structural mapping. Crowd-sourced annotation may lack expertise for nuanced attributes.

Open Question 4: What configuration density enables smooth navigation? Can hybrid pipelines combining computer vision for basic features with expert annotation for nuanced attributes maintain quality while scaling efficiently?

6 CONCLUSION

Digital fashion currently forces users to act as translators, converting vague aesthetic feelings into rigid professional terminology. Our Dual-layer framework argues that this Vocabulary Gap is an architectural flaw, not just an interface issue. By separating perceptual exploration from structural retrieval, we allow users to navigate high-dimensional spaces with precision, expressing preferences they recognize but cannot speak. Simultaneously, designers gain access to authentic preference data derived from behavior rather than surveys. While challenges regarding culture, interpretability, and scale remain, the core position stands: making the affect to structure logic transparent and spatially navigable is the key to truly intuitive style discovery.

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REFERENCES

- [1] A. Baldrati, D. Morelli, G. Cartella, M. Cornia, M. Bertini, and R. Cucchiara. Multimodal garment designer: Human-centric latent diffusion models for fashion image editing. *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 23336–23345, 2023. doi: 10.1109/iccv51070.2023.02138 2
- [2] R. Boardman and H. McCormick. Attention and behaviour on fashion retail websites: an eye-tracking study. *Inf. Technol. People*, 35:2219–2240, 2021. doi: 10.1108/itp-08-2020-0580 1
- [3] T. F. Boka, Z. Niu, and R. B. Neupane. A survey of sequential recommendation systems: Techniques, evaluation, and future directions. *Information Systems*, 125:102427, 2024. doi: 10.1016/j.is.2024.102427 4
- [4] D. Chen and P. Cheng. The style design of professional female vest based on kansei engineering. *International Journal of Clothing Science and Technology*, 2019. doi: 10.1108/ijest-07-2018-0090 2
- [5] M. Gan and M. Li. Design and realization of a collocation recommendation system for women’s clothing. *J Text Res*, 41:122–131, 2020. 3
- [6] X. Gu, Y. Wong, L. Shou, P. Peng, G. Chen, and M. Kankanalli. Multi-modal and multi-domain embedding learning for fashion retrieval and analysis. *IEEE Transactions on Multimedia*, 21:1524–1537, 2019. doi: 10.1109/tmm.2018.2876822 2, 3

- [7] R. Guidotti, A. Monreale, F. Giannotti, D. Pedreschi, S. Ruggieri, and F. Turini. Factual and counterfactual explanations for black box decision making. *IEEE Intelligent Systems*, 34:14–23, 2019. doi: 10.1109/mis.2019.2957223 [2](#)
- [8] Z. Guo, Z. Zhu, Y. Li, S. Cao, H. Chen, and G. Wang. Ai assisted fashion design: A review. *IEEE Access*, 11:88403–88415, 2023. doi: 10.1109/access.2023.3306235 [2](#), [3](#)
- [9] S. Hajahmadi, I. Calvi, E. Stacchiotti, P. Cascarano, and G. Marfia. Heritage elements and artificial intelligence as storytelling tools for virtual retail environments. *Digital Applications in Archaeology and Cultural Heritage*, 34:e00368, 2024. [1](#)
- [10] C. He, D. Parra, and K. Verbert. Interactive recommender systems: A survey of the state of the art and future research challenges and opportunities. *Expert Systems with Applications*, 56:9–27, 2016. doi: 10.1016/j.eswa.2016.02.013 [3](#)
- [11] M. Hu and Y. Zhong. Fashion style analysis and evaluation based on few-shot learning. *Wool Textile Journal*, 49(4), 2021. [2](#)
- [12] X. Huang, J. Li, J. Wu, J. Chang, D. Liu, and K. Zhu. Flexibly utilizing syntactic knowledge in aspect-based sentiment analysis. *Information Processing & Management*, 61(3):103630, 2024. doi: 10.1016/j.ipm.2023.103630 [3](#)
- [13] J. Jang, E. Hwang, and S.-H. Park. Lost your style? navigating with semantic-level approach for text-to-outfit retrieval. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pp. 8066–8075, 2024. [2](#)
- [14] H. Jiang and B. Ma. Style similarity algorithm based on clothing style. *Journal of Textile Research*, 2021. [2](#)
- [15] N. Kannakaklang, W. Thamviset, and S. Wongthanavas. Bigcan: A novel srs-based bidirectional graph convolution attention network for dynamic user preference and next-item recommendation. *Expert Systems with Applications*, 265:126016, 2025. doi: 10.1016/j.eswa.2024.126016 [4](#)
- [16] A. Karjus, M. C. Solà, T. Ohm, S. E. Ahnert, and M. Schich. Compression ensembles quantify aesthetic complexity and the evolution of visual art. *EPJ Data Science*, 12(1):21, 2023. [3](#)
- [17] W. Li. Research on clothing matching recommendation based on consumer electronic wardrobe. Master's thesis, Donghua University, Shanghai, 2025. (in Chinese). [2](#), [3](#)
- [18] Y. Li, R. Huang, and A. Dong. Fashion style recognition based on an improved bilinear-cnn. *Journal of Donghua University (Natural Science Edition)*, 47(3), 2021. [2](#)
- [19] W. Lian, K.-C. Wang, Y. Li, H.-Y. Chen, and C.-H. Yang. Affective-blue design methodology for product design based on integral kansei engineering. *Mathematical Problems in Engineering*, 2022. doi: 10.1155/2022/5019588 [3](#)
- [20] F. Lin, S. Liu, C. Zhang, J. Fan, and Z. Wu. Stylebert: Text-audio sentiment analysis with bi-directional style enhancement. *Information systems*, 114:102147, 2023. [3](#)
- [21] L. McInnes, J. Healy, and J. Melville. Umap: Uniform manifold approximation and projection for dimension reduction. *arXiv preprint arXiv:1802.03426*, 2018. [3](#)
- [22] G. Moro, S. Salvatori, and G. Frisoni. Efficient text-image semantic search: A multi-modal vision-language approach for fashion retrieval. *Neurocomputing*, 538:126196, 2023. doi: 10.1016/j.neucom.2023.03.057 [2](#)
- [23] C. Moruzzi and S. Margarido. A user-centered framework for human-ai co-creativity. *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*, 2024. doi: 10.1145/3613905.3650929 [2](#)
- [24] T. Pietilä. Negotiating aesthetic and emotional labor: Taste work by black fashion designers for their clients in johannesburg. *Anthropological Quarterly*, 98:123 – 151, 2025. doi: 10.1353/anq.2025.a957905 [2](#)
- [25] H. Quan, S. Li, and J. Hu. Product innovation design based on deep learning and kansei engineering. *Applied Sciences*, 2018. doi: 10.3390/app8122397 [1](#)
- [26] J. Ramos, H. A. Rahmani, X. Wang, X. Fu, and A. Lipani. Transparent and scrutable recommendations using natural language user profiles. In *Annual Meeting of the Association for Computational Linguistics*, 2024. [3](#)
- [27] R. Shimizu, T. Nakamura, and M. Goto. Partial visual-semantic embedding: Fine-grained outfit image representation with massive volumes of tags via angular-based contrastive learning. *Knowl. Based Syst.*, 277:110791, 2023. doi: 10.1016/j.knosys.2023.110791 [3](#)
- [28] R. Shimizu, Y. Saito, M. Matsutani, and M. Goto. Fashion intelligence system: An outfit interpretation utilizing images and rich abstract tags. *Expert Syst. Appl.*, 213:119167, 2022. doi: 10.1016/j.eswa.2022.119167 [1](#), [2](#)
- [29] G. Vallasciani, L. Stacchio, P. Cascarano, and G. Marfia. Creaixr: fostering creativity with generative ai in xr environments. In *2024 IEEE International Conference on Metaverse Computing, Networking, and Applications (MetaCom)*, pp. 1–8. IEEE, 2024. [2](#)
- [30] L. Xia, C. Huang, Y. Xu, and J. Pei. Multi-behavior sequential recommendation with temporal graph transformer. *IEEE Transactions on Knowledge and Data Engineering*, 35(6):6099–6112, 2023. doi: 10.1109/TKDE.2022.3175094 [4](#)
- [31] B. Yang. Research on apparel retrieval based on semantic similarity. Master's thesis, Beijing University of Posts and Telecommunications, Beijing, 2022. (in Chinese). [2](#)
- [32] X. Yue, C. Zhang, H. Fujita, and Y. Lv. Clothing fashion style recognition with design issue graph. *Applied Intelligence*, 51(6):3548–3560, 2021. [2](#)
- [33] Y. Zhang. Construction of garment style database based on fuzzy algorithm. *Wool Textile Journal*, 51(12):85–91, 2023. [2](#)
- [34] Y. Zhang, K. He, and R. Song. Image multi-feature fusion for clothing style classification. *IEEE Access*, 11:107843–107854, 2023. doi: 10.1109/access.2023.3320270 [1](#)